Indian Railway Delay Genome

Mapping failure propagation across the Indian railway network — which junctions, when delayed, infect the most other trains?

Total Trains	Unique Junctions	High-Risk Train Pairs	Most Dangerous Junction
8,366	30	46	Itarsi Jn

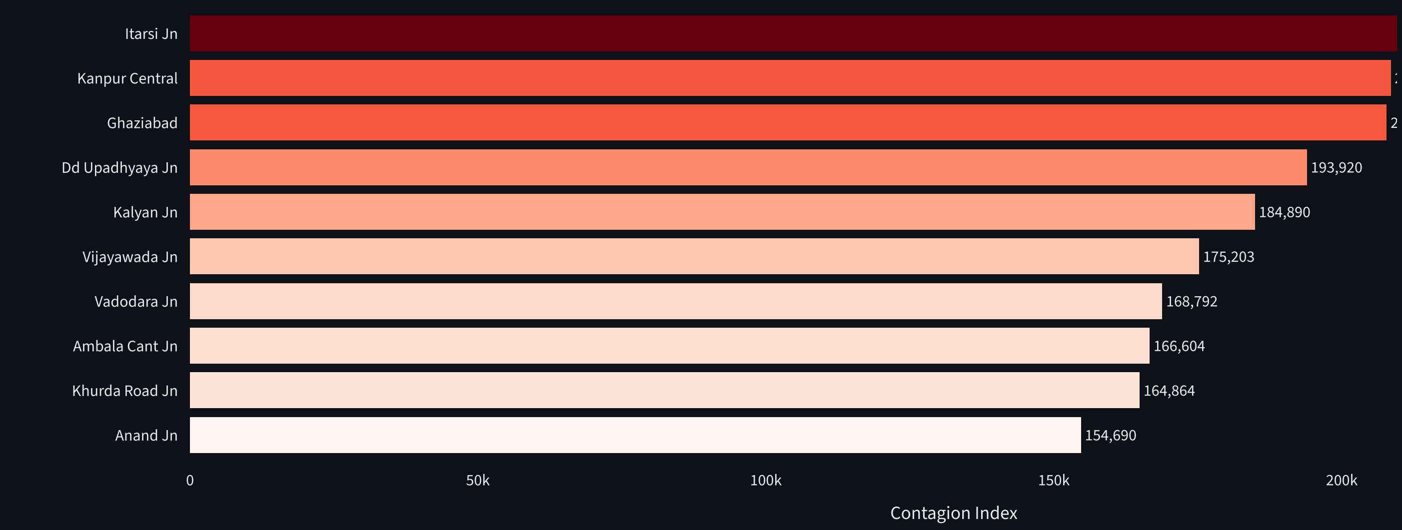
Contagion Index | Cascade Depth | High-Risk Pairs | Train Explorer

Junction Contagion Index

The **Contagion Index** measures how much damage a junction causes when delayed. It combines the number of trains passing through, the number of at-risk train pairs, and how many downstream stations the delay ripples to.

Show top N junctions

10



What this means

Itarsi Jn is the most dangerous junction in the network. **382** trains pass through it, creating **72,771** at-risk train pairs. A delay there can ripple across **1,323** downstream stations — nearly **18%** of all stations in India.

Junction	Trains Through	At-Risk Pairs	Downstream Stations
Itarsi Jn	382	72771	1
Kanpur Central	327	53301	1
Ghaziabad	317	50086	1
Dd Upadhyaya Jn	321	51360	1
Kalyan Jn	312	48516	1
Vijayawada Jn	310	47895	1
Vadodara Jn	313	48828	1
Ambala Cant Jn	288	41328	1
Khurda Road Jn	323	52003	1
Anand Jn	311	48205	

Indian Railway Delay Genome | Data: Indian Railways timetable (Kaggle) | Graph analysis: Neo4j AuraDB | Built with Streamlit + Plotly

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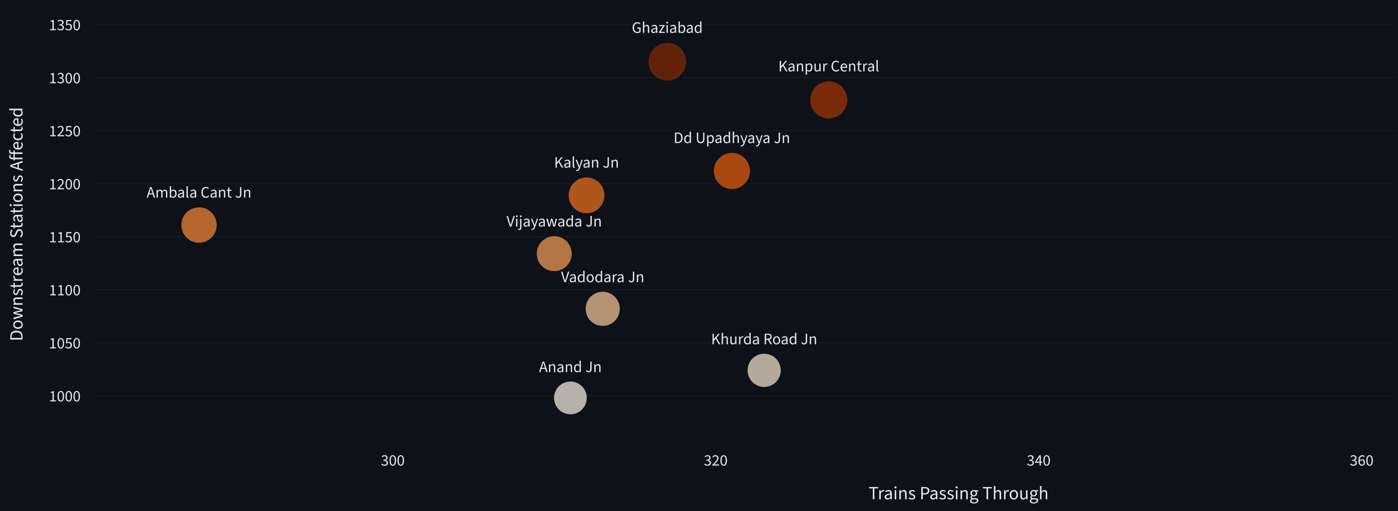
Cascade Depth — How Far Does a Delay Travel?

For each top junction, how many other stations are reachable through trains that pass through it? A delay at the junction travels with every train that was there.



Key insight: Ghaziabad has fewer trains than Itarsi but reaches almost as many downstream stations — it punches above its weight because it sits on multiple high-density corridors simultaneously.

Max downstream reach

1,323 stations

↑ via Itarsi Jn

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High-Risk Train Pairs

These are trains scheduled to arrive at the **same junction within 60 minutes** of each other. When one is delayed, it directly blocks the other's path.

Risk level

- ALL
- HIGH (≤20 min)
- MEDIUM (21–60 min)

HIGH risk pairs

46

MEDIUM risk pairs

72

Filter by junction

All

Junction	Train 1	Arr 1	Train 2	Arr 2	Gap
Itarsi Jn	SHC AMRIT BHARAT Train	23:45	APDJ AMRITBHARAT Train	23:45	
Kanpur Central	STP LTT SF EXP	00:02	POORVA EXPRESS Train	00:02	
Khurda Road Jn	SCL TVC EXPRESS Train	07:55	GHY SMVB EXP Train	07:55	
Khurda Road Jn	AGTL HUMSAFAR Train	10:25	CHZ SCL SF EXP Train	10:25	
Khurda Road Jn	SCL TVC EXPRESS Train	07:55	SCL SC EXPRESS Train	07:55	
Kanpur Central	STP LTT SF EXP	00:02	PBH BPL EXPRESS Train	00:02	
Khurda Road Jn	GHY SMVB EXP Train	07:55	SCL SC EXPRESS Train	07:55	
Khurda Road Jn	GUWAHATI EXP	01:15	CBE SCL SF EXP	01:15	
Kanpur Central	LTT SITAPUR EXP Train	13:40	UDYOG NAGRI EXP Train	13:40	
Khurda Road Jn	SCL TVC EXPRESS Train	07:55	SCL CBE EXPRESS Train	07:55	

Note: Pairs with 0-minute gap are structurally problematic — they are scheduled to arrive simultaneously, meaning one always waits.

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🏆 Contagion Index | Cascade Depth | ⚠️ High-Risk Pairs | 🔍 Train Explorer

Train Route Explorer

Look up any train's full route and see which high-risk junctions it passes through.

Search by train name or number

e.g. Rajdhani or 12951

Type a train name or number above to explore its route.

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